# Command

Undoable editor keystrokes

Season 3 · GoF · Beginner

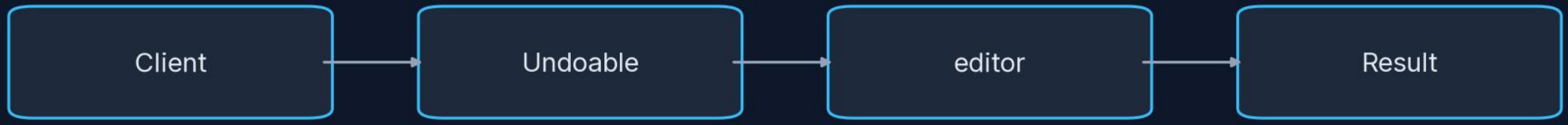
# Today's Hook

- Problem: Undoable editor keystrokes
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Command with a runnable Demo.
- Failure path is locked by ProblemTest.

# The Problem

- Scenario: Undoable editor keystrokes
- Constraints: solve «Undoable editor keystrokes» with Command — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Undoable editor keystrokes
- Success looks like: Hi

# Structure




Command: Undoable editor keystrokes

# Real-World Scenario

- Running example: Undoable editor keystrokes
- Key types: Editor, Command, WriteCommand, CommandBus
- Happy-path stdout cue: Hi
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

# Naive Code

```java
package com.buildwithwaqar.patterns.gof.behavioral.command.problem;

import java.util.*;

public final class NaiveEditor {
  private String text="";
  public void write(String s){ text+=s; }
  public void undo(){ /* cannot undo */ }
  public String text(){ return text; }
}
```

# Why Naive Hurts

- Naive version fails to cleanly solve: Undoable editor keystrokes
- Without Command, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

# Solution

```
package com.buildwithwaqar.patterns.gof.behavioral.command.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;

final class Editor { private final StringBuilder sb=new StringBuilder();
  void write(String s){ sb.append(s); } void deleteLast(int n){ sb.delete(Math.max(0,sb.length()-n), sb.length()); }
  String text(){ return sb.toString(); }
}
interface Command { void execute(); void undo(); }
final class WriteCommand implements Command {
  private final Editor ed; private final String chunk;
  WriteCommand(Editor ed, String chunk){ this.ed=ed; this.chunk=chunk; }
  public void execute(){ ed.write(chunk); } public void undo(){ ed.deleteLast(chunk.length()); }
}
final class CommandBus {
  private final Deque<Command> hist=new ArrayDeque<>();
  void run(Command c){ c.execute(); hist.push(c); }
  void undo(){ if(!hist.isEmpty()) hist.pop().undo(); }
}
```

# Runnable main

```
package com.buildwithwaqar.patterns.gof.behavioral.command.problem;

import java.util.*;
import java.util.concurrent.*;

public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== Command: undoable editor ===");
    System.out.println("--- naive ---");
var n=new NaiveEditor(); n.write("Hi"); n.undo(); System.out.println(n.text());
System.out.println("--- command ---");
var ed=new Editor(); var bus=new CommandBus();
bus.run(new WriteCommand(ed,"Hello")); bus.run(new WriteCommand(ed,"!"));
System.out.println(ed.text()); bus.undo(); System.out.println("afterUndo="+ed.text());
  }
}
```

# Walkthrough Output

```
=== Command: undoable editor ===
--- naive ---
Hi
--- command ---
Hello!
afterUndo=Hello
```

# Step-by-Step Trace

- Match stdout to stages — look for: Hi
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp ... 16-command ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

# Modern Java 25 / Zazr

```
// Java 25 direction for Command
// Prefer records + sealed where they shrink the demo.
// Problem: Undoable editor keystrokes
```

# Trade-offs

- Use Command when «Undoable editor keystrokes» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Command in product language.

# Related Patterns

- Name the pattern usually confused with Command and separate the forces.
- Sketch the roles for «Undoable editor keystrokes» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

# Recap

- Problem: Undoable editor keystrokes
- Naive pain → Command boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.